

Project Planning

Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits

Project Objective

To develop an interactive Tableau dashboard and story - embedded in a Flask web application - that analyzes food ordering behaviour, cuisine and meal-type preferences, city-wise demand, and delivery performance. The project transforms raw order data into visual insights that support business, operations, and customer-facing decisions.

Project Scope

- Collecting and preparing the food ordering dataset for analysis.
- Cleaning and transforming data (Order ID, Order Value, Delivery Fee, Cuisine, Meal Type, City, Time Taken for Delivery).
- Creating key performance indicators (KPIs).
- Designing interactive dashboards with charts and filters.
- Developing a Tableau Story to present business insights.
- Performing calculation-field-driven and filter-based performance testing.
- Embedding the dashboard and story into a Flask web application.
- Generating documentation and a final report.

Project Schedule

Phase 1: Planning

- Define project objectives.
- Identify stakeholders (Business Analyst, Operations Manager, Customer).
- Gather requirements.
- Prepare the project plan.

Phase 2: Data Preparation

- Collect and import the dataset.
- Clean and preprocess the data.
- Validate data quality.
- Organize fields for analysis.

Phase 3: Dashboard Development

- Design KPI cards.
- Create charts and graphs.
- Add filters and interactive features.
- Arrange dashboard layout for responsiveness.

Phase 4: Story Development

- Build a Tableau Story.
- Present key business questions.
- Highlight insights on demand, cuisine, and delivery.

- Summarize findings.

Phase 5: Performance Testing

- Verify dashboard accuracy and load performance.
- Test filters and interactions.
- Validate calculated fields.
- Confirm visualization count and responsiveness.

Phase 6: Web Integration

- Publish the dashboard and story to Tableau Public.
- Build the Flask application (Home, Dashboard, Story, About).
- Embed the dashboard and story using the tableau-viz component.

Phase 7: Documentation and Submission

- Prepare the final report.
- Capture dashboard/story screenshots.
- Record the project demonstration video.
- Submit the complete project.

Resources Required

Software

- Tableau Desktop / Tableau Public
- Microsoft Excel
- Python 3 with Flask
- Code editor (VS Code)
- Web browser

Hardware

- Windows 10/11 computer
- Minimum 4 GB RAM
- Intel Core i3 processor or above

Deliverables

- Cleaned dataset
- Tableau Workbook
- Interactive Dashboard
- Tableau Story
- Flask web application with embedded dashboard/story
- Project documentation (phase-wise)
- Final report
- Project demonstration video

Risk Management

Risk	Mitigation Strategy
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Missing or inconsistent data	Perform data cleaning and validation before analysis
Dashboard performance issues	Optimize calculated fields and limit unnecessary visuals
Incorrect KPI calculations	Test and verify every calculated field manually
Tableau embed not loading in Flask	Use the official tableau-viz component and test across browsers
Delays in project completion	Follow the phase-wise schedule and review progress regularly

Conclusion

A well-structured project plan ensures the successful development of the Food Ordering Behaviour and Consumer Trends dashboard. By following each phase systematically, the project delivers accurate visual analytics - accessible through a responsive Flask web app - that help businesses understand customer behaviour, monitor delivery performance, and make data-driven decisions.